Name: _____

Instructions: All solutions should be prepared carefully and neatly. Any questions that ask you to explain or justify should have answers written in complete sentences. All solution sets shall be completed on this packet and handed in at the beginning of class on the due date below. Careless presentation (e.g. bad handwriting, pen scribbles, doodles, wasted space, etc) will result in a deduction of points at my discretion. If the question asks to submit a graph or plot, be sure to print the plot and include it in your final lab report. Use a PENCIL and if you make a mistake, use an eraser. Each student should prepare their own solutions that demonstrate their own comprehension of the material. Submitted work that does not demonstrate clearly the process by which one arrived at the answer will not receive credit of any kind. Academic dishonesty will not be tolerated.

Desmos Lab II

MATH 182 – Calculus II

Due: Friday, October 17

Read: Section 7.6

Problem Number	Available Points	Your Points
1	14	
Total	14	

1. Numerical Integration: Suppose a region is bounded by the functions $y = e^{x^2}$, $x = 0$, $x = 1$ and the x-axis. We will attempt to find its area using the definite integral and various numerical approximation techniques.

   (a) Create your account on the Desmos website so you can save your created graphs. After completing this lab, please save your created Desmos graph and share its Desmos link below.

   (b) Plot and shade the region on an appropriate window in Desmos.

   (c) What is the definite integral to find the area? What is the area (according to Desmos)? Are you able to solve this definite integral manually? Explain.

   (d) Let's try to approximate the area using the Trapezoidal Rule with $n = 4$. What is your result? How close is it to its real value?

   (e) Let's try to approximate the area using Simpson's Rule with $n = 4$. What is your result? How close is it to its real value?

   (f) Based on the previous results from (d) and (e), which rule gives you a closer estimation?

(g) Let's try to approximate the area using the Trapezoidal Rule with $n = 6$. What is your result? How close is it to its real value?

(h) How does the change of the n value impact the accuracy of the Trapezoidal Rule?

(i) Let's try to approximate the area using Simpson's Rule with $n = 6$. What is your result? How close is it to its real value?

(j) How does the change of the n value impact the accuracy of Simpson's Rule?

(k) Based on the previous results from (g) and (i), which rule gives you a closer estimation?